

HDOC DAY-14 REPORT

Menu-Driven C Program Using Switch-Case

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Activity	HDOC Day-14
Batch	15
Language	C

1. Problem Statement

Prepare the algorithm and test case table. Then write, compile, execute, and test a C program against the prepared test cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Check whether all digits of the number are unique.
2. Find the frequency of each digit from 0 to 9.

Activity Instruction

HDOC Day-14



Summarize this email

Dear Student,
The following are the **HDOC Day-14** activities for Batches 15 & 16:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
 - **E.g. 1:** Number: 12345, Output: Unique
 - **E.g. 2:** Number: 12234, Output: Not Unique
- Find the frequency of each digit from `0` to `9`.
 - **E.g.** Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

Warm Regards.

2. Algorithm

Step 1: Start.

Step 2: Display the menu: 1) Check whether all digits are unique, 2) Find frequency of each digit from 0 to 9.

Step 3: Read the user's choice.

Step 4: Read a positive integer n.

Step 5: Use switch(choice) to perform the selected operation.

Step 6: For case 1, initialize freq[0] to freq[9] to zero. Extract each digit using temp % 10, increment its frequency, and remove the digit using temp / 10.

Step 7: Check the frequency array. If any digit has frequency greater than 1, print Not Unique; otherwise print Unique.

Step 8: For case 2, initialize the frequency array to zero, extract every digit and count its occurrence.

Step 9: Print freq_0 through freq_9.

Step 10: For any other menu choice, print Invalid choice!.

Step 11: Stop.

3. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Choice = 1 Number = 12345	Unique	Unique	Pass
2	Choice = 1 Number = 12234	Not Unique	Not Unique	Pass
3	Choice = 2 Number = 112333	0:0, 1:2, 2:1, 3:3, 4-9:0	0:0, 1:2, 2:1, 3:3, 4-9:0	Pass
4	Choice = 12234 Number = 12234	Invalid choice!	Invalid choice!	Pass

4. Program Code

The following screenshots show the C program used for the activity.

```
UW PIC0 5.09 File: day14.cpp

#include <stdio.h>

int main()
{
    int choice, n, temp, digit;
    int freq[10];
    int i, unique;

    printf("MENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit from 0 to 9\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    switch (choice)
    {
        case 1:
            for (i = 0; i < 10; i++)
                freq[i] = 0;

            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            unique = 1;
            for (i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }

            if (unique == 1)
                printf("Output: Unique\n");
            else
                printf("Output: Not Unique\n");

            break;

        case 2:
            for (i = 0; i < 10; i++)
                freq[i] = 0;

            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            printf("Digit Frequencies:\n");

            for (i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

```
UW PIC0 5.09 File: day14.cpp

            digit = temp % 10;
            freq[digit]++;
            temp = temp / 10;
        }

        unique = 1;
        for (i = 0; i < 10; i++)
        {
            if (freq[i] > 1)
            {
                unique = 0;
                break;
            }
        }

        if (unique == 1)
            printf("Output: Unique\n");
        else
            printf("Output: Not Unique\n");

        break;

    case 2:
        for (i = 0; i < 10; i++)
            freq[i] = 0;

        temp = n;

        while (temp > 0)
        {
            digit = temp % 10;
            freq[digit]++;
            temp = temp / 10;
        }

        printf("Digit Frequencies:\n");

        for (i = 0; i < 10; i++)
        {
            printf("freq_%d = %d\n", i, freq[i]);
        }

        break;

    default:
        printf("Invalid choice\n");
    }

    return 0;
}
```

5. Compilation, Execution and Testing

The program was compiled and executed in the terminal and tested using the prepared cases.

```
dhruvpandey@DHUVs-MacBook-Air ~ % clang day14.c -o day14
clang: error: no such file or directory: 'day14.c'
clang: error: no input files
dhruvpandey@DHUVs-MacBook-Air ~ % clang day14.c -o day14
clang: error: no such file or directory: 'day14.c'
clang: error: no input files
dhruvpandey@DHUVs-MacBook-Air ~ % nano day14.c
dhruvpandey@DHUVs-MacBook-Air ~ % clang day14.c -oday14
dhruvpandey@DHUVs-MacBook-Air ~ % ./day14
MENU
1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1
Enter a positive integer: 12345
Output: Unique
dhruvpandey@DHUVs-MacBook-Air ~ % 2
zsh: command not found: 2
dhruvpandey@DHUVs-MacBook-Air ~ % ./day14
MENU
1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1
Enter a positive integer: 12234
Output: Not Unique
dhruvpandey@DHUVs-MacBook-Air ~ % ./day14
MENU
1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 2
Enter a positive integer: 112333
Digit Frequencies:
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
dhruvpandey@DHUVs-MacBook-Air ~ % ./day14
MENU
1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 12234
Enter a positive integer: 12234
Invalid choice!
dhruvpandey@DHUVs-MacBook-Air ~ %
```

6. Result

The menu-driven C program was successfully compiled and executed. The program correctly identifies unique and repeated digits, displays the frequency of digits from 0 to 9, and handles an invalid menu choice.